

Context/Purpose:

I'm building a website to serve as my visualization portfolio for presenting to future employers. I want this website to be unique to me and incorporate some of the things I love along with the projects and visualizations I've done throughout my years as a student. I am currently a Master's at Wageningen University in GIS, focusing on Remote Sensing and Machine Learning.

Contents:

- Picture of me :) this is attached and I want this to be visible without scrolling
- Interactive world map feature:
 - I want this to be a subtle colored map following the color pallet of the website
 - At each location I want a little bubble (colored as one of the reference colors) that is hoverable and displays some info about that place and time of my life as described below.
 - I've lived in:
 - Seattle, WA, USA
 - This is where I was born and grew up (2002) I love the mountains, sea and overall wildlife of this area
 - Bremen, DE
 - 2019-2020: I participated in rotary exchange here when I was 16-17 and lived in host families. I was inspired to move abroad by my parents: my grew up in switzerland and my dad sailed around the world after college for a few years.
 - From my mom I know german and during this exchange I got to really practice my speaking german
 - Florence, IT - 2021
 - After I graduated high school I didn't know what I wanted to do but I wanted to be adventurous after being cooped up inside from COVID.
 - I lived in Florence for a semester with VERTO (US program) where I took college courses
 - Statistics, Art History, Sociology
 - Turrialba, Cost Rica - 2022
 - Also with VERTO I did another semester in costa rica
 - I began learning spanish which I'm still taking classes in right now
 - Was living at an agricultural research institute taking classes in ecology and comparative govt and politics, and spanish
 - I was infatuated by the nature on campus and of the country and traveled all around
 - Prague, CZ
 - Figured out what I liked and did a bachelor's in environmental data science at CZU in prague
 - I took ecology, hydrology, data management, visualization, and GIS courses - especially enjoyed the GIS courses

- Did my thesis in flood and ET dynamics and analyzed the 1997 flood in CZ and what/where the main sources were.
- Wageningen, NL
 - Currently a Master's student in Geo-Information Sciences and love it!
 - Took courses in Geoscripting, Geotools, Remote Sensing, Advanced Remote Sensing, Machine Learning, Deep Learning, and Remote Sensing for Atmospheric Composition
 - Student consultancy project for Antea Group working on automating their obstacle detection for airports, working with Airborne Lidar data (AHN).
 - Really into radar, climate change monitoring, wildfire sensing, urban heat, and making the world a better place for all 😊
- 4 boxes with my past projects that just include the title and part of the visualization.
 - These are all clickable and go to pages that contain more information including links to the storymap (GIS in context) and website prototype (IDHV).
 - <https://storymaps.arcgis.com/stories/8bfadb89de5b4b60bb6b278223f3c407>
 - <https://www.figma.com/proto/DQQdoqkyrZlihxVoRdX7qW/Hotspotters---Vote-for-Watchtower?node-id=780-1074&t=7HrFu7tyk1SgPv79-0&scaling=min-zoom&content-scaling=fixed&page-id=0%3A1&starting-point-node-id=794%3A115>

Past projects to highlight:

- GIS in context
 - “Scotland's Peatlands: Opportunities for Investors”
 - Why Peatland Restoration Matters
 - Peat is a carbon-rich organic material formed over millennia in waterlogged ecosystems like bogs. Peatlands act as natural carbon sinks, storing vast amounts of CO₂. When degraded, they release emissions, accelerating climate change. Restoration can reverse this, offering scalable climate and biodiversity benefits. As can be seen in the map on the left, peatland is widespread across the Scottish landscape but is more prevalent in the northern regions. This map shows different levels of peatland coverage, where dark blue indicates areas of pure peat, and light blue represents regions with a mix of peat and regular soil. Peatland is a key feature of the Scottish natural landscape, covering around 20% of the country. Together these soils store around 1.6 million Tonnes of carbon, which is more than in all UK's forests combined (Scotland's soils, 2025). Healthy peatlands also play a vital role in flood regulation and water quality, and they host unique biodiversity. However, the intensive use of these landscapes has resulted in land degradation on a large scale. Estimates suggest that 80% of Scottish peatlands are currently degraded (Scottish government, 2021). Instead of capturing and storing

carbon, degraded peatlands emit carbon back into the atmosphere in the order of 6 Mt CO₂e/year (Cabinet Secretary for Rural Affairs, Land Reform and Islands, 2023).

- Was storymap showcasing Scotland's peatlands and the need for more investors because there isn't enough money from the government. It tells of why peatlands are important and how our audience (investors) can help them from degrading. We mapped peat depth and areas in selected study areas and calculated the carbon being stored in shallow vs deep peat.
- Geoscripting
 - Modeling Elephant Presence Using Environmental Covariates in the Welgevonden Game Reserve
 - Objective: This project aims to model elephant presence based on environmental covariates, visualizing the results in an interactive dashboard.
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 - Specifically, we aim to answer these questions: * Where do elephants go most frequently go to throughout the years 2018-2020 in the Welgevonden Game Reserve? * Using a logistic regression model, how accurately can we predict the event of an elephant being at a location given a specific time and temperature? * Which environmental factors are most strongly correlated with elephant presence in the study area? * Can kernel density estimates (KDE) of elephant locations accurately identify hotspots of elephant activity?
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 - Although this project uses elephant data from the Welgevonden Game Reserve, users can easily adapt the code for their own reserve by changing the data inputs and environmental variables.
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 - Methods
 - Data Sources
 - All data used in this project belongs to the Welgevonden Game Reserve, and shared with Wageningen University for education purposes only.
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 - This data includes elephant tracking data, elephant metadata, weather data, and additional GIS layers used as environmental covariates.
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 - Data Summary
 - Elephant tracking data: 10 African elephants (linked with tag attribute) in the reserve from 2018-2021, with temporal resolution mainly between 30 and 10 minutes
 - Elephant metadata: Includes the tag, sex, and name of all elephants
 - Weather data: Captured near the main gate (-24.203323°, 27.903202°), with 3 minute time intervals between 01-01-2018 and 01-01-2020.
 - GIS Layers:
 - Human Settlements: 2 vector point layers depicting staff accommodations and lodges.

- Roads: A vector point layer showing the roads on the reserve.
- Elevation: A DEM raster layer of the elevation of the reserve.
- Reserve Area: Boundary polygon shapefile of the reserve.
- Reservoirs and Cribs: A vector point layer showing artificial water sources.
- Logistic Regression Model
- Logistic regression to predict elephant presence (1) or absence (0).
- - Uses environmental covariates (e.g., distance to roads, settlements, elevation, temperature, time).
- - Evaluated with confusion matrix, ROC curve, and AUC; multicollinearity checked with VIF.
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- Output: Pre-trained model saved as final_model.rds and used in the dashboard for interactive probability maps.
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- Results
- Summary of model performance (accuracy, p-values, etc.)
- Example outputs: plots, raster maps
- Screenshots of the interactive dashboard (optional)
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- Allows users to:
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- View KDE heatmaps of elephant locations
- Adjust environmental conditions: temperature and time
- Outputs a map showing either KDE or predicted probability of elephant presence
- How to run
- Train the model
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- Run train_model.R to prepare the data, train the logistic regression model, and save the trained model as an .rds file (final_model.rds).
- This step only needs to be done once! The saved model will be used directly by the dashboard.
- Launch the dashboard
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- Run main.R to start the interactive Shiny application.
- The dashboard will load the saved model (final_model.rds) and have options of displaying:
- A Kernel Density Estimation (KDE) map of elephant locations.
- A Location Predictor tool where users can adjust environmental sliders (temperature, time of day) to visualize predicted elephant occurrence probabilities.
- File Structure
- main.R: * Prepares and then runs dashboard.
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- train_model.R: * Trains the model based on the data given in config.R, and saves the final model (models/final_model.rds).
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- ElephantFunctions: * Contains all the functions to process the data, train the model, and run the model from the dashboard.
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- server.R: * Contains code to run the dashboard.
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- ui.R: * Specifies dashboard functionality and user interaction.
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- config.R: * Contains code for loading packages, functions, and defines data paths.
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- prepare_data.R * Reads and runs functions to prepare data for training the model, as well as running the model.
- Geotools
 - Created a cost raster to calculate the best route given many constraints between wastewater areas and the largest water refinement basin
 - All in ArcGIS
 - Deliverables (in one map format) - this map is attached
 - 1. The largest water refinement basin & industrial wastewater areas
 - 2. The calculated pipelines of all possible connections (from the wastewater areas to the largest refinement basin in Druten)
 - 3. The final suitability cost raster (Expected values are between 1 and 100)
- IDHV Groenlo project
 - Output: A Figma prototype, a clickable website that was a voting platform for 3 wildlife watchtower locations in Groenlo, NL. We wanted to showcase the wildlife activity and viewing activity at each location so the public could vote on their favorite location and make it a reality.
 - Flew a drone with a heat sensor early in the morning to capture wildlife activity
 - Found birds and deer
 - Output for the website was an orthomosaic heatmap of each of the three locations, each with labeled animals.
 - Conducted a viewshed analysis using a TLS (terrestrial lidar scanner) point cloud data at the three locations. This is shown in a simple user friendly map on the website, showcasing in 2D what can be seen.

Design

- Color palette:
 - --color-1: #C8CEEE;
 - --color-2: #777A9D;
 - --color-3: #9C7894;

- --color-4: #55597C;
- --color-5: #EEE7EF;
- --color-6: #DDB5B6;
- --color-7: #BCA8C1;
- --color-8: #FEE2D7;

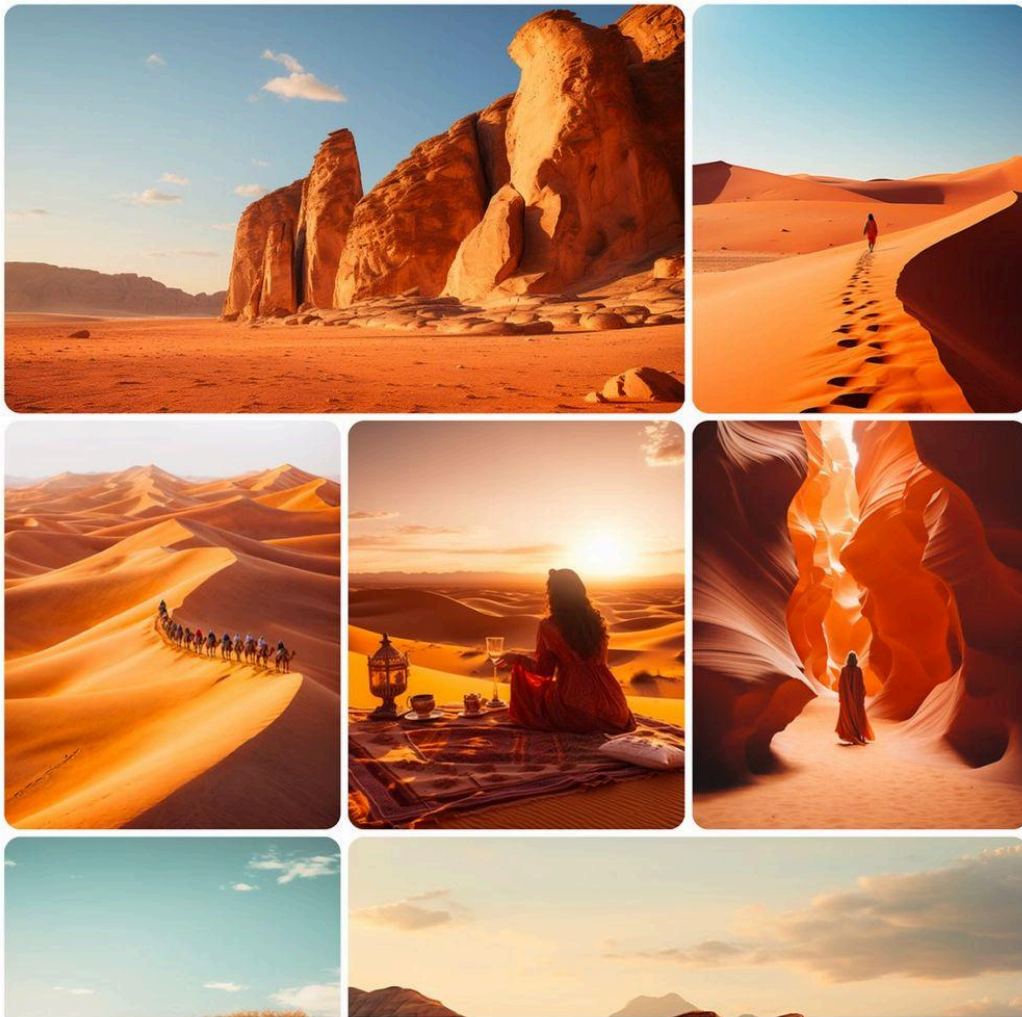
Content Order:

- Header
 - Should just contain the titles of everything in the home page so that this immediately scrolls smoothly down to that section location

Our Stories

Photo Gallery

Captured moments from our
desert trips and scenic routes.



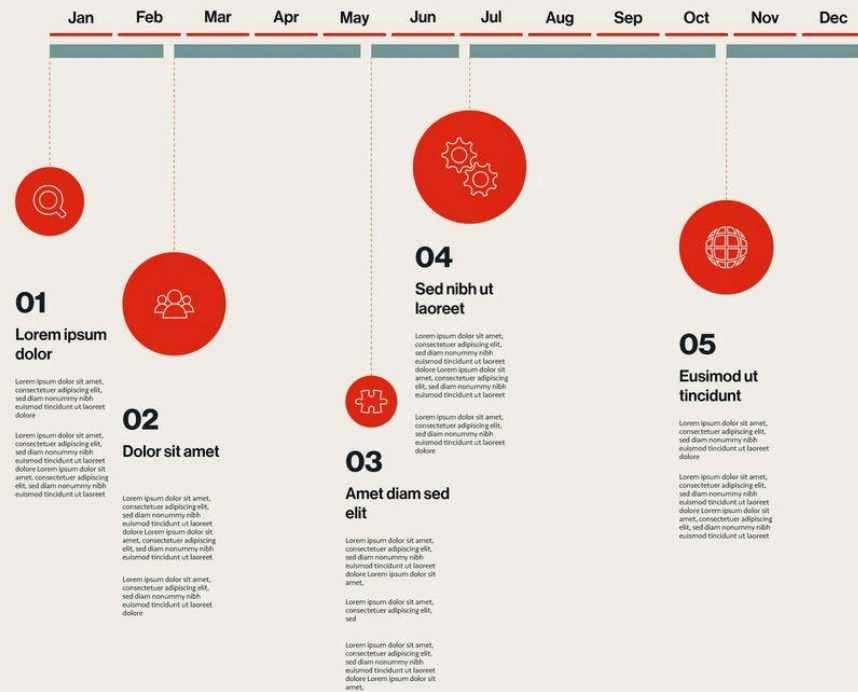
- I like these fonts for title:
 - Instead of “Photo Gallery” I want “Hi, I’m Sophia”
 - Geneve - [link](#)



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- Below this I want a few sentences describing me and my ambitions professionally. Just write a couple lines and I will update this later. Shouldn't span the entire width of the page.
- Like the bubbles in the reference layout photo - I want the left top larger bubble to a nature photo and the right smaller one to be a photo of me.
 - Mountains.JPG
 - Me_pointing_left.png
 -
- Then the map as described above (world_map.png)
 - Png map can be as reference and I'd love it to be colored with one of the colors in the palette - I can change this later too
 - The map should take up the whole width of the page in one of the box bubbles
- Below the map should be a horizontal timeline stating my education/work history going from Pedalheads job (left) to present master's study (right)

Project Steps

INFOGRAPHICS TEMPLATE



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- This is an example of the type of timeline I want. I like the bubbles and use of colors (should be in my color palette).
- The months should be switched to years
- This should overall act as a simple CV where it includes the date, school/workplace and a short description for work or thesis
 - Ex. CZU in Prague, Bachelor's in environmental data science 2022-2025
 - Again this is all in CV
- Then I want to showcase my 4 projects described above and in 2x2 bubble format
 - Each bubble should have:
 - Course title and project title
 - In bubble should be the photo given in the past_visualizations folder
- After this I want a title "Skills", and then a list of skills (in CV)
 - QGIS, ArcGIS, Python, R, SQL, GEE
 - LiDAR, DL, ML, radar, multispectral, hyperspectral, remote sensing, atmospheric gas retrievals, RTM

- Contact:
 - Section with linkedin link, email, and button to download my CV
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